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REGENERATION OF SUPERFICIAL DAMAGE IN THE SCLERACTINIAN CORALS *AGARICIA AGARICITES* F. *PURPUREA* AND *PORITES ASTREOIDES*

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ABSTRACT

We studied regeneration of artificial lesions (simulating damage by predators and physical factors) on the living surfaces of the corals *Agaricia agaricites* f. *purpurea* and *Porites astreoides in situ* at 25 and 18–23 m, respectively. *Porites* regenerated ≈50% of 1 cm^2 lesions completely, *Agaricia* less than 10%. Larger, 5 cm^2, lesions were not re-covered by either species in the 140 days of observation. The results demonstrate the vulnerability of some corals to mechanical damage and illustrate species-specific characteristics.

The living surfaces of stony corals (Scleractinia) are relatively free of alien organisms. Parts of these surfaces are known to be damaged and removed through predation, sedimentation and other causes. The resulting free space on the coral skeleton is available for organisms that compete for this resource. The invading organisms are a potential threat to the survival of the coral colony. Spatial competitors may grow over the remaining living coral tissue, and excavating organisms, such as clionid sponges, may penetrate the coral skeleton. Whether a coral is able to re-cover a lesion in the living surface, depends both on the invading organisms and the regenerating capacity of the coral colony. The literature on regeneration in corals concerns skeletal regeneration (e.g. Wood Jones, 1907; Stephenson and Stephenson, 1933) and few data are available on regeneration of the living tissue.

To investigate the impact of superficial damage on stony corals, we have studied regeneration of two types of artificial lesions of two sizes on several coral species and at various depths. Here we report on our experiments with *Agaricia agaricites* (Linnaeus) forma *purpurea* and *Porites astreoides* Lamarck.

MATERIALS AND METHODS

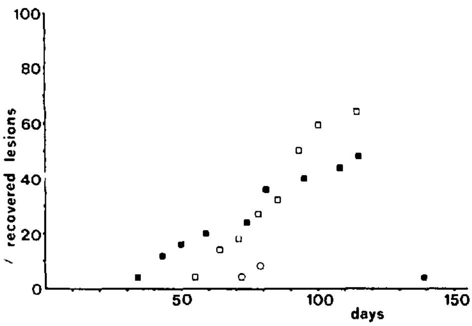
Our experimental colonies were located at a fringing reef 700–800 m northwest of the entrance of Piscadera Bay, southwest coast of Curaçao. We selected colonies of *A. agaricites* and *P. astreoides* at depths of 25 and 18–23 m, respectively. These species are numerically and functionally important on the study reef (Bak, 1977; Bak and Engel, 1979).

The experimental lesions were designed to represent two types of natural damage. The first type consists of removal of living tissue only (by predation such as gastropods, e.g. *Coralliophila abbreviata* [Lamarck], the polychaete *Hermodice carunculata* [Pallas] and through sedimentation). The second type includes removal of living tissue as well as the underlying superficial skeletal elements (through grazing activities of fishes, Scaridae, and *Diadema antillarum* Philippi).

Between September 1976 and March 1977 we made three series of circular lesions on different colonies of each species (method: Bak et al., 1977). The lesions were all on the horizontal plane. One series consisted of 1 cm^2 tissue lesions, a second series comprised 1 cm^2 tissue and skeleton lesions and a third series 5 cm^2 tissue and skeleton lesions. All lesions (total of 147) were surrounded, in radial directions, by at least 4 cm of living tissue. During the ≈140 days of observation after infliction of the lesions, the lesions were checked weekly for the size of the lesion, nature of colonizing organisms and tissue regeneration characteristics. Eleven control colonies of each species were checked for significant natural changes on the living surfaces.

RESULTS

There was a striking difference in the regeneration capacity of the two species (Fig. 1). In *Porites astreoides* 64% of the tissue lesions and 48% of the tissue and



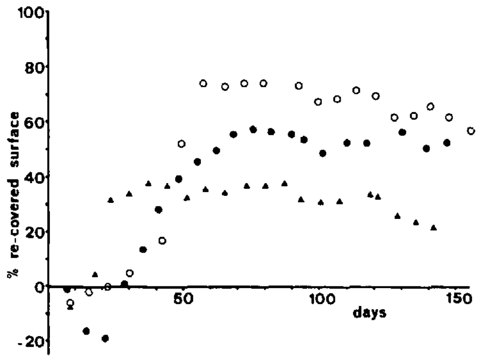


Figure 1. (Left) Rates of regeneration of the two coral species expressed as percentage of lesions completely recovered. Open circles, *A. agaricites* 1 cm^2 tissue lesions, n = 25; closed circles, *A. agaricites* 1 cm^2 tissue and skeleton lesions, n = 26; open squares, *P. astreoides* 1 cm^2 tissue lesions, n = 22; closed squares, *P. astreoides* 1 cm^2 tissue and skeleton lesions, n = 25. Figure 2. (Right) Regeneration in *A. agaricites* expressed as mean surface area of all lesions. Triangles, 5 cm^2 tissue and skeleton lesions, n = 24. For other symbols see Fig. 1.

skeleton lesions were completely regenerated within 140 days. In *Agaricia agaricites* only 8% of the tissue lesions and 4% of the tissue and skeleton lesions were 100% regenerated. None of the 5 cm^2 (tissue and skeleton) lesions was wholly re-covered by coral tissue in either species.

Only few data became available on the regeneration rates of tissue versus tissue and skeleton lesions in *Agaricia agaricites* (Fig. 1). But these, as well as comparison of the total regenerated surface (all lesions) of the different series (Fig. 2), show that *A. agaricites* was more successful in covering tissue lesions than in regenerating tissue and skeleton lesions. In *Porites astreoides* tissue and skeleton lesions are the first to be completely regenerated (Fig. 1). This difference between regeneration rates of the two types of lesions is later obscured, when tissue lesions are fast regenerating and re-cover much of the damaged lesions (Figs. 1, 3).

Figures 2, 3 show that after infliction of the lesions, the surrounding tissue retracts, resulting in an initial enlargement of the lesions. This was followed by a phase of successful regeneration, subsequently succeeded by a period of stabilization. The stabilization, in terms of area covered, indicates a decrease in regeneration capacity which is also illustrated by the decrease (through time) in the number of lesions re-covering $\geq$75% of the damaged surface (Fig. 4).

We measured the size of the experimental colonies to correlate this with the percentage surface of the lesions covered after 70 and 140 days, respectively. Colony size ranged from 150–1,100 cm^2 (1 cm^2 lesion series) and from 250–1,350 cm^2 (5 cm^2 lesion series). No linear correlations were found between size and regeneration rate in either species (product-moment correlation, $P > 0.1$).

The two species employ different regeneration strategies. In *Agaricia* the regenerating lip of tissue, followed by newly calcified skeletal material, forms a dome-shaped roof over the original lesions (Bak et al., 1977). In *Porites* small, originally colourless, polyps are formed, behind the regenerating lip which surrounds the center of the lesion. Cross sections of regenerated lesions showed that the regenerating surface is completely in contact with the surface of the lesion.

The only organisms colonizing the lesions in abundance were filamentous algae. Algae became conspicuous, 3–4 weeks after infliction of the lesions, on the sed-

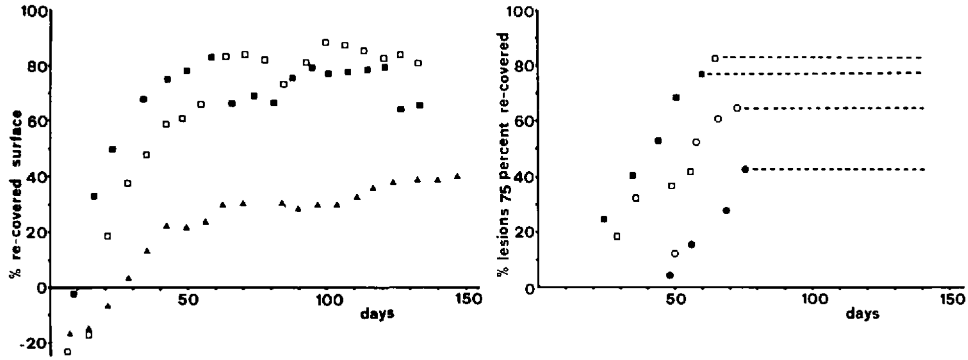


Figure 3. (Left) Regeneration in *P. astreoides* expressed as mean surface area of all lesions. Triangles, 5 cm^2 tissue and skeleton lesions, n = 25. For other symbols see Fig. 1. Figure 4. (Right) Regeneration expressed as percentage of lesions regenerating ≥75% of surface area. After ≈70 days the number of lesions re-covering for 75% does not increase. This is indicated by level of broken lines. For explanation of symbols see Fig. 1.

iment patches that covered the scars. Other colonizers were observed only on the 5 cm^2 lesions. They included burrowing sponges (4 occurrences) and the anemone *Heteractis lucida* Duchassaing and Michelotti (2 occurrences).

Table 1 shows the changes in the living surface area of the control colonies.

Discussion

The most capable regenerator of surface lesions of the various coral species we studied so far at different depths was *Montastrea annularis* (8–12 m, Bak et al., 1977), followed by *Agaricia agaricites* (8–12 m, Bak et al., 1977) and *Porites astreoides* (18–23 m), while *A. agaricites* (25 m) had the lowest regenerative ability. The lower regeneration capacity of *A. agaricites* in relatively deeper water is surprising regarding the higher calcification rate of this, and other, coral species in deeper water (Barnes and Taylor, 1973; Bak, 1976; Erez, 1978). Although superficially damaged coral showed slower skeletal growth (*Acropora palmata* Lamarck, Bak, 1976 p. 303; Bak, in prep.) it is possible that calcification rates *per se* are no indication of regeneration capacity.

The low competitive potential of *Agaricia* is confirmed by the inability shown by the control colonies of the species to maintain the integrity of the living surface (Table 1). Bak and Engel (1979) found that *A. agaricites* is by far the most abundant juvenile settler on the reefs and suggest that *A. agaricites* is an opportunistic species (*sensu* Grassle and Grassle, 1974), channelling more energy into reproductive effort than into maintenance.

As in shallower water *Agaricia agaricites* is more capable of regenerating tissue

Table 1. Changes in size (area) of natural lesions on control colonies after a 140 days period

	Area <50% (increased-decreased)		Area Increased >50%		Area Decreased >50%	
	No.	%	No.	%	No.	%
A. agaricites (n = 11)	7	64	4	36	0	0
P. astreoides (n = 11)	9	82	0	0	2	18

lesions than tissue and skeleton lesions. In *Porites astreoides* damage to the skeleton appears to accelerate the regeneration processes initially. The slopes of the regression lines for the two types of lesions (Fig. 1, tissue/time, and skeleton/time, regression coefficients 1.098 and 0.528, respectively) differ statistically significantly (student's t test, $P < 0.05$), but we are not sure of the biological significance. In *Porites* the tissue is not limited to the polypary surface layer but includes a subsurface system of coenenchymal tubes lined by calicoblastic epidermis (Roos, 1967). This subsurface tissue was removed neither in the tissue lesions nor in the tissue and skeleton ones. We recorded the behavior of the regenerating edge of the lesions and do not know if these remnants, under the sediment patches, facilitated regeneration.

The inability of both *A. agaricites* and *P. astreoides* to re-cover 5 cm^2 lesions shows the size of the damaged area to be a major parameter in regeneration.

Connell's (1973) and Loya's (1976) data showed a relatively low regeneration capacity in small (<80 cm^2), respectively very small ($\leqslant 8$ cm^2) colonies. Our data (Bak et al., 1977; this paper) do not suggest a simple linear relation between colony size and regeneration capacity. A complex of parameters such as type and size of lesions, species and size of colony may prevent such a general correlation. The size of the colony may be important in that a minimal area is needed to enable regeneration, while after this threshold area is available colony size becomes less important.

Although colonization of the fresh lesions was noticeably slow, compared with shallow water (Bak et al., 1977), all lesions were covered with dense algal-sediment mats before any lesion was wholly re-covered. It appears that colonization of a lesion *per se* does not prevent complete regeneration.

The data show (Figs. 2, 3) that after a 70-day period, an equillibrium is established between the algae-covered surface of the lesions and the regenerating coral in terms of area covered. Although lesions are still completely closed after this period (Fig. 1), this complete regeneration is restricted to lesions that are regenerated for more than 75% in the first 70 days (Fig. 4). If a lesion is not nearly completely re-covered by the regenerating coral in the first period this will result in a long-lasting dead spot on the coral surface prone to colonization by other spatial competitors and excavating organisms. A decisive factor in this process is not the presence of sediment and filamentous algae, but a physiological characteristic of the coral colony, such as a decrease in certain stimuli or resources.

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